定理 22.4 设 P,Q,R 是定义在光滑曲面 $S:z=z(x,y),(x,y)\in D$ 上的连续函数，以 S 的上侧为正侧，则

$$\iint_S P(x,y,z)\,dydz + Q(x,y,z)\,dzdx + R(x,y,z)\,dxdy$$

$$= \iint_D (P(x,y,z(x,y))(-z_x) + Q(x,y,z(x,y))(-z_y) + R(x,y,z(x,y)))\,dxdy.$$

证 由于

$$\cos\alpha = \frac{-z_x}{\sqrt{1+z_x^2+z_y^2}},\quad \cos\beta = \frac{-z_y}{\sqrt{1+z_x^2+z_y^2}},\quad \cos\gamma = \frac{1}{\sqrt{1+z_x^2+z_y^2}},\quad dS = \sqrt{1+z_x^2+z_y^2}\,dxdy,$$

因此

$$\iint_S P(x,y,z)\,dydz + Q(x,y,z)\,dzdx + R(x,y,z)\,dxdy$$

$$= \iint_S (P(x,y,z)\cos\alpha + Q(x,y,z)\cos\beta + R(x,y,z)\cos\gamma)\,dS$$

$$= \iint_D (P(x,y,z(x,y))(-z_x) + Q(x,y,z(x,y))(-z_y) + R(x,y,z(x,y)))\,dxdy.$$

□

例 3　计算 $\iint_S (2x+z)\,dydz + z\,dxdy$，其中 $S = \{(x,y,z) \mid z = x^2 + y^2, z \in [0, 1]\}$，取上侧.

解　$z_x = 2x, z_y = 2y$,

$$I = \iint_D ((2x + x^2 + y^2)(-2x) + x^2 + y^2)\,dxdy = \iint_D (y^2 - 3x^2)\,dxdy$$

$$= -\iint_D 2x^2\,dxdy = -2\int_0^{2\pi} d\theta \int_0^1 r^2 \cos^2\theta\, r\,dr = -\frac{\pi}{2},$$

其中由于 $x(x^2 + y^2)$ 是 x 的奇函数，$\iint_D x(x^2 + y^2)\,dxdy = 0$；又由对称性，

$$\iint_D x^2\,dxdy = \iint_D y^2\,dxdy.$$

□

12.9 Gauss公式与Stokes公式

Green 公式 —推广→ Gauss 公式
Stokes公式

一、向量微分算子

二、Gauss公式

三、Stokes公式

一、向量微分算子

1. 梯度算子: $\nabla = \frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k}$

又称为∇(Nabla)算子, 或哈密尔顿(Hamilton) 算子.

设$u = u(x, y, z)$, 则

$$\nabla u = \frac{\partial u}{\partial x}\vec{i} + \frac{\partial u}{\partial y}\vec{j} + \frac{\partial u}{\partial z}\vec{k} = \operatorname{grad} u$$

$$\nabla^2 u = \nabla \cdot \nabla u = \nabla \cdot \operatorname{grad} u$$

$$= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \Delta u$$

with $c_1, c_2 > 0$ and $c_3 := c_2 \int_0^\infty (1+\sigma^{-\mu}) e^{-\mu} d\sigma < \infty$ due to $\mu \in (\frac{1}{2}, 1)$, here we used (3.8) in Lemma 3.2, and that for all $t \in (0, T_{\max})$,

$$
\begin{aligned}
&\int_0^t \|B^\mu e^{-(t-s)B} u(\cdot,s)\nabla v(\cdot,s)\|_{L^k(\Omega)} ds \\
&\le c_4 \int_0^t \left(1 + (t-s)^{-\mu-\frac{3}{2}\left(\frac{1}{l}-\frac{1}{k}\right)}\right) e^{-(t-s)} \|u(\cdot,s)\nabla v(\cdot,s)\|_{L^l(\Omega)} ds \\
&\le c_4 \int_0^t \left(1 + (t-s)^{-\mu-\frac{3}{2}\left(\frac{1}{l}-\frac{1}{k}\right)}\right) e^{-(t-s)} \|u(\cdot,s)\|_{L^l(\Omega)} \|\nabla v(\cdot,s)\|_{L^\infty(\Omega)} ds \\
&\le c_4 \int_0^t \left(1 + (t-s)^{-\mu-\frac{3}{2}\left(\frac{1}{l}-\frac{1}{k}\right)}\right) e^{-(t-s)} \|u(\cdot,s)\|_{L^l(\Omega)} \qquad (3.18) \\
&\quad \times \left(\|\nabla(v(\cdot,t) - e^{-tB} v_0)\|_{L^\infty(\Omega)} + \|\nabla e^{-tB} v_0)\|_{L^\infty(\Omega)}\right) ds \\
&\le c_4 K_2 H_p^{\frac{p}{p-1}\cdot\left(\frac{l-3}{3l}+\delta\right)}(t) \int_0^t \left(1 + (t-s)^{-\mu-\frac{3}{2}\left(\frac{1}{l}-\frac{1}{k}\right)}\right) e^{-(t-s)} ds
\end{aligned}
$$

2. 散度

设 $\vec{A} = P(x,y,z)\vec{i} + Q(x,y,z)\vec{j} + R(x,y,z)\vec{k},$

则 $$\mathrm{div}\vec{A} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

$$= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \cdot (P, Q, R)$$

$$= \nabla \cdot \vec{A}$$

性质： $$\mathrm{div}(k_1\vec{A}_1 + k_2\vec{A}_2) = k_1\mathrm{div}\vec{A}_1 + k_2\mathrm{div}\vec{A}_2$$

$$\mathrm{div}(u\vec{A}) = u\mathrm{div}\vec{A} + \mathrm{grad}u \cdot \vec{A}$$

3. 旋度

$$\vec{A} = P(x,y,z)\vec{i} + Q(x,y,z)\vec{j} + R(x,y,z)\vec{k},$$

$$\mathrm{rot}\vec{A} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \nabla \times \vec{A}$$

性质： $\mathrm{rot}(k_1\vec{A}_1 + k_2\vec{A}_2) = k_1\mathrm{rot}\vec{A}_1 + k_2\mathrm{rot}\vec{A}_2$

场论中的三个重要概念

设 $u=u(x,y,z)$, $\vec{A}=(P,Q,R)$, $\nabla=\left(\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial z}\right)$, 则

梯度: $\operatorname{grad} u=\left(\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial u}{\partial z}\right)=\nabla u$

散度: $\operatorname{div}\vec{A}=\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}=\nabla\cdot\vec{A}$

旋度: $\operatorname{rot}\vec{A}=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ P & Q & R\end{vmatrix}=\nabla\times\vec{A}$

思考与练习

设 $r=\sqrt{x^2+y^2+z^2}$, 则

$$\operatorname{div}(\operatorname{grad} r)=\underline{\quad 2/r \quad};\ \operatorname{rot}(\operatorname{grad} r)=\underline{\quad \vec{0} \quad}.$$

提示： $\operatorname{grad} r=\left(\frac{x}{r},\frac{y}{r},\frac{z}{r}\right)$

$$\frac{\partial}{\partial x}\left(\frac{x}{r}\right)=\frac{r-x\cdot x/r}{r^2}=\frac{r^2-x^2}{r^3},\qquad \frac{\partial}{\partial y}\left(\frac{y}{r}\right)=\frac{r^2-y^2}{r^3}$$

$$\frac{\partial}{\partial z}\left(\frac{z}{r}\right)=\frac{r^2-z^2}{r^3}$$

三式相加即得 $\operatorname{div}(\operatorname{grad} r)$

$$\operatorname{rot}(\operatorname{grad} r)=\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ \dfrac{x}{r} & \dfrac{y}{r} & \dfrac{z}{r} \end{vmatrix}=(0,0,0)$$

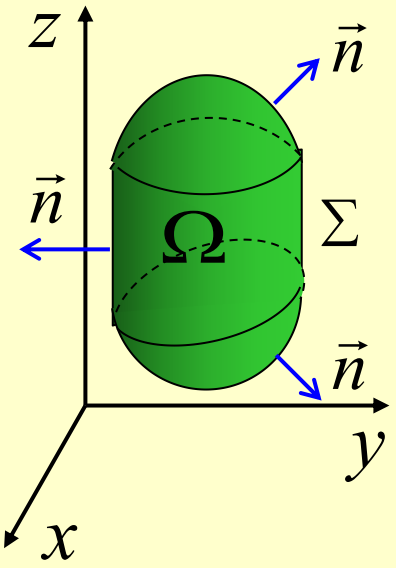
二、高斯 (Gauss) 公式



定理1. 设空间闭区域 Ω 由分片光滑的**闭曲面** Σ 所围成, Σ 的方向**取外侧**, 函数 P, Q, R 在 Ω 上有连续的一阶偏导数, 则有

$$\oiint_{\Sigma} P\,\mathrm{d}y\,\mathrm{d}z + Q\,\mathrm{d}z\,\mathrm{d}x + R\,\mathrm{d}x\,\mathrm{d}y$$

$$= \iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$$



或 $$\iiint_{\Omega} \nabla \cdot \vec{A}\,\mathrm{d}v = \oiint_{\Sigma} \vec{A} \cdot \overrightarrow{\mathrm{d}S}$$ **(Gauss 公式)**

Pf: 先证 $\oint\limits_{\Sigma_{外}} R(x,y,z)\,dxdy = \iiint\limits_{\Omega} \frac{\partial R(x,y,z)}{\partial z}\,dv$

(1) 当 Ω 为 xy-型区域　　Ω: $z_1(x,y) \le z \le z_2(x,y)$, $(x,y)\in D_{xy}$

$z_1(x,y)$ 为下曲面，$z_2(x,y)$ 为上曲面.

Σ_3: $F(x,y)=0$ 在曲面上点 $M(x,y,z)$　　$\vec{n} = \pm\{F_x, F_y, 0\}$

$\vec{n^0} = \pm\left\{\frac{F_x}{\sqrt{F_x^2+F_y^2}}, \frac{F_y}{\sqrt{F_x^2+F_y^2}}, 0\right\}$ 则 $\cos\gamma = 0 \Rightarrow \gamma = \frac{\pi}{2}$.

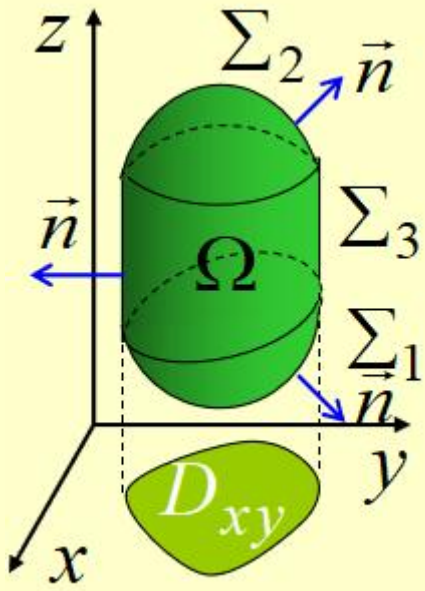
Σ_2: $z = z_2(x,y)$, $(x,y)\in D_{xy}$, γ 全锐

Σ_1: $z = z_1(x,y)$, $(x,y)\in D_{xy}$, γ 全钝

$$\iiint\limits_{\Omega} \frac{\partial R}{\partial z}dv = \iint\limits_{D_{xy}} dxdy \int_{z_1(x,y)}^{z_2(x,y)} \frac{\partial R(x,y,z)}{\partial z}dz$$

$$= \iint\limits_{D_{xy}} \left(R(x,y,z)\Big|_{z_1(x,y)}^{z_2(x,y)} \right) dxdy = \iint\limits_{D_{xy}} [R(x,y,z_2(x,y)) - R(x,y,z_1(x,y))]\,dxdy$$

$$= \iint\limits_{D_{xy}} [R(x,y,z_2(x,y)) - R(x,y,z_1(x,y))]\,d\sigma.$$



$$\oint_{\Sigma外} R(x,y,z)\,dx\,dy = \iint_{\Sigma_2+\Sigma_1+\Sigma_3} R(x,y,z)\,dx\,dy.$$

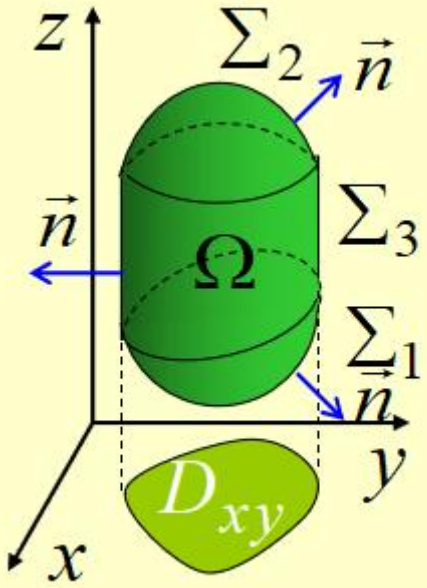
$$= \iint_{D_{xy}} R(x,y,z_2(x,y))\,d\sigma + \iint_{D_{xy}} R(x,y,z_1(x,y))\cdot(-1)\,d\sigma$$

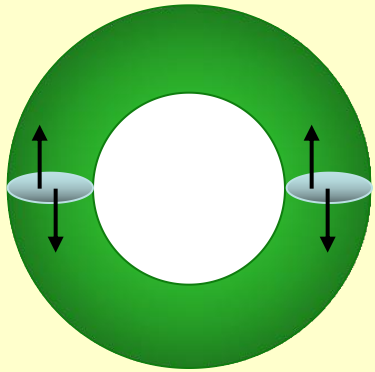
$$= \iint_{D_{xy}} [R(x,y,z_2(x,y)) - R(x,y,z_1(x,y))]\,d\sigma$$

$$\therefore \oint_{\Sigma外} R(x,y,z)\,dx\,dy = \iiint_{\Omega} \frac{\partial R}{\partial z}\,dv$$

同理 $$\oint_{\Sigma外} P(x,y,z)\,dy\,dz = \iiint_{\Omega} \frac{\partial P}{\partial x}\,dv$$

$$\oint_{\Sigma外} Q(x,y,z)\,dz\,dx = \iiint_{\Omega} \frac{\partial Q}{\partial y}\,dv.$$





若 Ω 不是 XY–型区域 , 则可引进辅助面将其分割成若干个 XY–型区域, 在辅助面正反两侧面积分正负抵消, 故上式仍成立 .

类似可证 $\iiint_{\Omega}\frac{\partial P}{\partial x}\mathrm{d}x\mathrm{d}y\mathrm{d}z=\oiint_{\Sigma}P\mathrm{d}y\mathrm{d}z$

$$\iiint_{\Omega}\frac{\partial Q}{\partial y}\mathrm{d}x\mathrm{d}y\mathrm{d}z=\oiint_{\Sigma}Q\mathrm{d}z\mathrm{d}x$$

三式相加, 即得所证 Gauss 公式:

$$\iiint_{\Omega}\left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}\right)\mathrm{d}x\mathrm{d}y\mathrm{d}z$$

$$=\oiint_{\Sigma}P\mathrm{d}y\mathrm{d}z+Q\mathrm{d}z\mathrm{d}x+R\mathrm{d}x\mathrm{d}y$$

若高斯公式中 $P=x, Q=y, R=z$，则有

$$\iiint_V (1+1+1)\,dxdydz = \oiint_S x\,dydz + y\,dzdx + z\,dxdy.$$

于是得到应用第二型曲面积分计算空间区域 V 的体积公式

$$\Delta V = \frac{1}{3}\oiint_S x\,dydz + y\,dzdx + z\,dxdy.$$

例：计算 $\oiint_{\Sigma外} x^3dydz + y^3dzdx + z^3dxdy$　$\Sigma: x^2+y^2+z^2=R^2$ 外侧

解：由高斯公式，原式 $= \iiint_\Omega (3x^2+3y^2+3z^2)dV$

$$= 3\int_0^{2\pi}d\theta \int_0^{\pi}d\varphi \int_0^{R} \rho^2 \cdot \rho^2\sin\varphi\, d\rho$$

$$= 3\int_0^{2\pi}d\theta \cdot \int_0^{\pi}\sin\varphi\, d\varphi \cdot \int_0^{R}\rho^4 d\rho$$

$$= 3\cdot 2\pi \cdot 2 \cdot \frac{R^5}{5} = 3\cdot\frac{4\pi}{5}R^5 = \frac{12\pi}{5}R^5$$

变式 1 求 $I=\oiint_S \dfrac{x^3\mathrm{d}y\mathrm{d}z+y^3\mathrm{d}z\mathrm{d}x+z^3\mathrm{d}x\mathrm{d}y}{\sqrt{x^2+y^2+z^2}}$，$S$ 是球面

$x^2+y^2+z^2=a^2(a>0)$ 外侧.

解 $I=\dfrac{1}{a}\oiint_S x^3\mathrm{d}y\mathrm{d}z+y^3\mathrm{d}z\mathrm{d}x+z^3\mathrm{d}x\mathrm{d}y=\dfrac{12}{5}\pi a^4$.

S 是球面 $x^2+y^2+z^2=a^2(a>0)$ 外侧，$\oiint_S x^3\mathrm{d}y\mathrm{d}z+y^3\mathrm{d}z\mathrm{d}x+z^3\mathrm{d}x\mathrm{d}y=\dfrac{12}{5}\pi a^5$.

变式 2 求 $I=\oiint_S x\mathrm{d}y\mathrm{d}z+y\mathrm{d}z\mathrm{d}x+z\mathrm{d}x\mathrm{d}y$，$S$ 是球面

$x^2+y^2+z^2=a^2(a>0)$ 外侧.

解 $I=3\iiint_V \mathrm{d}v\ (V: x^2+y^2+z^2\le a^2)$

$=4\pi a^3$.

变式 3 求 $I=\oiint_S x^3\mathrm{d}y\mathrm{d}z+y^2\mathrm{d}z\mathrm{d}x+z\mathrm{d}x\mathrm{d}y$，$S$ 是球面

$x^2+y^2+z^2=a^2(a>0)$ 外侧.

解 $I=\iiint_V(3x^2+2y+1)\mathrm{d}v\,(V:x^2+y^2+z^2\le a^2)$

$$=\iiint_V(x^2+y^2+z^2)\mathrm{d}v+2\iiint_V y\mathrm{d}v+\iiint_V \mathrm{d}v$$

$$=\frac{4}{5}\pi a^5+\frac{4}{3}\pi a^3.$$

例 1 求 $I=\oiint_S x^2\mathrm{d}y\mathrm{d}z+y^2\mathrm{d}z\mathrm{d}x+z^2\mathrm{d}x\mathrm{d}y$，$S$ 是正方体 V：

$0\le x\le a$，$0\le y\le a$，$0\le z\le a$ 的全表面外侧.

解 由 Gauss 公式

$$I=2\iiint_V(x+y+z)\mathrm{d}v=2(\bar{x}+\bar{y}+\bar{z})\iiint_V\mathrm{d}v$$

$$=2\left(\frac{a}{2}+\frac{a}{2}+\frac{a}{2}\right)\cdot a^3=3a^4.$$

1. Gauss 公式优势明显；

2. 一次式的重积分计算

（1）利用对称性

（2）利用重心公式

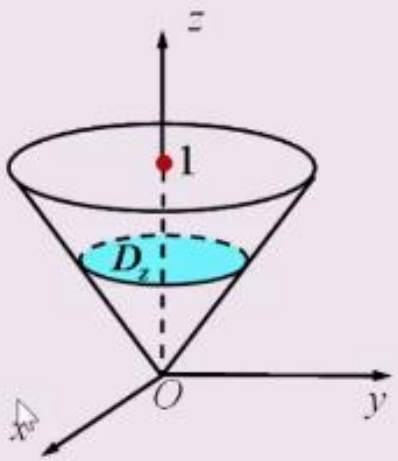
变式 1 求 $I=\oiint_S x^2\mathrm{d}y\mathrm{d}z+y^2\mathrm{d}z\mathrm{d}x+z^2\mathrm{d}x\mathrm{d}y$，$S$ 是球面

$x^2+y^2+z^2=a^2(a>0)$ 外侧.

解 $I=2\iiint_V(x+y+z)\mathrm{d}v\,(V:x^2+y^2+z^2\le a^2)$

$=0.$

变式2 求 $I=\oiint_S x^2\mathrm{d}y\mathrm{d}z+y^2\mathrm{d}z\mathrm{d}x+z^2\mathrm{d}x\mathrm{d}y$，$S$ 是锥面 $z=\sqrt{x^2+y^2}$ 与平面 $z=1$ 所围立体的表面外侧.



解 $I=2\iiint_V(x+y+z)\mathrm{d}v=2\iiint_V z\mathrm{d}v$

$$=2\int_0^1 z\mathrm{d}z\iint_{D_z}\mathrm{d}x\mathrm{d}y\,(D_z:x^2+y^2\le z^2)$$

$$=2\pi\int_0^1 z^3\mathrm{d}z=\frac{\pi}{2}.$$